```
!pip install transformers torch --upgrade
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.3.0)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transf
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from tr
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transf
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from transformers
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transfo
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-package
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torc
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from to
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torc
Requirement already satisfied: hf-xet<2.0.0,>=1.3.2 in /usr/local/lib/python3.12/dist-packages (fr
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from h
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from ji
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (fr
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from http
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.2
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (f
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdow
```

```
from transformers import pipeline
import torch

print("Transformers and Torch imported successfully!")
print("Torch version:", torch.__version__)
```

```
Transformers and Torch imported successfully!
Torch version: 2.10.0+cpu
```

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

model_name = "facebook/bart-large-cnn"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForSeq2SeqLM.from_pretrained(model_name)

text = """
Artificial intelligence is transforming industries across the world.
It helps companies analyze data quickly and make better decisions.
"""

# Encode the input text
inputs = tokenizer([text], max_length=1024, return_tensors="pt", truncation=True)

# Generate summary IDs directly from the model
summary_ids = model.generate(
    inputs["input_ids"],
    num_beams=4,
    max_length=50,
    min_length=20,
```

```
    early_stopping=True,
    no_repeat_ngram_size=3
)

# Decode the generated summary
summary_text = tokenizer.decode(summary_ids[0], skip_special_tokens=True)

print(summary_text)
```

```
Loading weights: 100%                                    511/511 [00:00<00:00, 7099.03it/s]
Artificial intelligence is transforming industries across the world. It helps companies analyze da
```

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

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    inputs["input_ids"],
    num_beams=4,
    max_length=50,
    min_length=20,
    early_stopping=True,
    no_repeat_ngram_size=3
)

# Decode the generated summary
summary_text = tokenizer.decode(summary_ids[0], skip_special_tokens=True)

print(summary_text)
```

```
Loading weights: 100%                                    511/511 [00:00<00:00, 2620.31it/s]
Artificial intelligence is transforming industries across the world. It helps companies analyze da
```

```
texts = [

"""
Artificial intelligence is transforming industries across the world.
Companies use machine learning algorithms to analyze huge volumes of
data and improve decision making. AI systems can identify patterns,
automate repetitive tasks, and increase productivity. However, the
rapid development of AI also raises concerns about ethics, privacy,
and job displacement. Experts emphasize responsible AI development
and proper regulations.
""",

"""
Climate change is a major global challenge. Rising temperatures,
melting glaciers, and increasing sea levels threaten ecosystems and
human communities. Governments around the world are investing in
renewable energy such as solar and wind power. Public awareness and
sustainable practices are necessary to reduce carbon emissions and
```

```
protect the environment for future generations.
""",

"""
Software engineering involves designing, developing, testing, and
maintaining software applications. Teams often follow agile
methodologies that encourage collaboration and iterative
development. Tools such as version control systems, automated
testing frameworks, and continuous integration pipelines help
developers build reliable and scalable software systems.
""",

"""
Once upon a time in a quiet village, a young girl loved exploring
the forests near her home. One day she discovered a hidden path
that led to a mysterious ancient tree. The villagers believed the
tree held magical powers and protected the forest. Curious and
brave, the girl began visiting the tree every day and learning the
secrets of nature.
""",

"""
Two colleagues were discussing their project during a meeting.
One of them was worried about the tight deadline and limited
resources. The other suggested dividing the project into smaller
tasks and prioritizing the most important features first. By
working together and communicating clearly, they developed a plan
to complete the project successfully.
"""
]
```

```
for i, t in enumerate(texts):

    # Encode the input text
    inputs = tokenizer([t], max_length=1024, return_tensors="pt", truncation=True)

    # Generate summary IDs directly from the model
    summary_ids = model.generate(
        inputs["input_ids"],
        num_beams=4,
        max_length=80,  # Using the max_length from the original intent
        min_length=25,  # Using the min_length from the original intent
        early_stopping=True,
        no_repeat_ngram_size=3
    )

    # Decode the generated summary
    summary_text = tokenizer.decode(summary_ids[0], skip_special_tokens=True)

    print("\n-------------------------------")
    print("TEXT", i+1)
    print("\nOriginal Text:\n", t)

    print("\nSummary:\n", summary_text)
```

```
-------------------------------
TEXT 1

Original Text:

Artificial intelligence is transforming industries across the world.
Companies use machine learning algorithms to analyze huge volumes of
data and improve decision making. AI systems can identify patterns,
automate repetitive tasks, and increase productivity. However, the
rapid development of AI also raises concerns about ethics, privacy,
and job displacement. Experts emphasize responsible AI development
```

```
and proper regulations.


Summary:
 Companies use machine learning algorithms to analyze huge volumes of data and improve decision

------------------------------
TEXT 2

Original Text:

Climate change is a major global challenge. Rising temperatures,
melting glaciers, and increasing sea levels threaten ecosystems and
human communities. Governments around the world are investing in
renewable energy such as solar and wind power. Public awareness and
sustainable practices are necessary to reduce carbon emissions and
protect the environment for future generations.


Summary:
 Climate change is a major global challenge. Rising temperatures,melting glaciers, and increasin

------------------------------
TEXT 3

Original Text:

Software engineering involves designing, developing, testing, and
maintaining software applications. Teams often follow agile
methodologies that encourage collaboration and iterative
development. Tools such as version control systems, automated
testing frameworks, and continuous integration pipelines help
developers build reliable and scalable software systems.


Summary:
 Software engineering involves designing, developing, testing, andmaintaining software applicati

------------------------------
TEXT 4

Original Text:

Once upon a time in a quiet village, a young girl loved exploring
```

```
# Encode the input text
inputs = tokenizer([texts[0]], max_length=1024, return_tensors="pt", truncation=True)

# Generate summary IDs directly from the model
summary_ids = model.generate(
    inputs["input_ids"],
    num_beams=4,
    max_length=30,
    min_length=10,
    early_stopping=True,
    no_repeat_ngram_size=3
)

# Decode the generated summary
short_summary_text = tokenizer.decode(summary_ids[0], skip_special_tokens=True)

print("Short Summary:")
print(short_summary_text)
```

```
Short Summary:
Companies use machine learning algorithms to analyze huge volumes of data. However, therapid devel
```

```
# Encode the input text
inputs = tokenizer([texts[0]], max_length=1024, return_tensors="pt", truncation=True)
```

```
# Generate summary IDs directly from the model
summary_ids = model.generate(
    inputs["input_ids"],
    num_beams=4,
    max_length=60,
    min_length=20,
    early_stopping=True,
    no_repeat_ngram_size=3
)

# Decode the generated summary
medium_summary_text = tokenizer.decode(summary_ids[0], skip_special_tokens=True)

print("Medium Summary:")
print(medium_summary_text)
```

```
Medium Summary:
Companies use machine learning algorithms to analyze huge volumes of data and improve decision mak
```

```
# Encode the input text
inputs = tokenizer([texts[0]], max_length=1024, return_tensors="pt", truncation=True)

# Generate summary IDs directly from the model
summary_ids = model.generate(
    inputs["input_ids"],
    num_beams=4,
    max_length=120,
    min_length=40,
    early_stopping=True,
    no_repeat_ngram_size=3
)

# Decode the generated summary
long_summary_text = tokenizer.decode(summary_ids[0], skip_special_tokens=True)

print("Long Summary:")
print(long_summary_text)
```

```
Long Summary:
Companies use machine learning algorithms to analyze huge volumes of data and improve decision mak
```

```
types = ["AI Article", "Climate News", "Technical Text", "Narrative Story", "Conversation"]

for i in range(len(texts)):

    # Encode the input text
    inputs = tokenizer([texts[i]], max_length=1024, return_tensors="pt", truncation=True)

    # Generate summary IDs directly from the model
    summary_ids = model.generate(
        inputs["input_ids"],
        num_beams=4,
        max_length=70,  # Using the max_length from the original intent
        min_length=25,  # Using the min_length from the original intent
        early_stopping=True,
        no_repeat_ngram_size=3
    )

    # Decode the generated summary
    summary_text = tokenizer.decode(summary_ids[0], skip_special_tokens=True)

    print("\n==============================")
    print("Text Type:", types[i])
    print("Summary:", summary_text)
```

```
==============================
Text Type: AI Article
Summary: Companies use machine learning algorithms to analyze huge volumes of data and improve dec

==============================
Text Type: Climate News
Summary: Climate change is a major global challenge. Rising temperatures,melting glaciers, and inc

==============================
Text Type: Technical Text
Summary: Software engineering involves designing, developing, testing, andmaintaining software app

==============================
Text Type: Narrative Story
Summary: A young girl discovered a hidden path that led to a mysterious ancient tree. The villager

==============================
Text Type: Conversation
Summary: Two colleagues were discussing their project during a meeting. One of them was worried ab
```

```
long_text = """
Artificial Intelligence has become a major area of research and
innovation. Organizations across industries are using AI to improve
efficiency and automate tasks. Machine learning models analyze large
datasets and detect patterns that humans cannot easily see.

In healthcare, AI helps doctors diagnose diseases and analyze medical
images. In finance, AI systems detect fraud and manage risks. Retail
companies use AI to study customer behavior and personalize marketing.

Despite its advantages, AI raises ethical concerns including bias,
privacy issues, and job automation. Governments and organizations are
developing guidelines to ensure responsible AI development.

Education systems are also adapting by introducing courses related
to data science, machine learning, and artificial intelligence.
As technology continues to evolve, balancing innovation with ethical
responsibility will be important for the future.
"""

# Encode the input text
inputs = tokenizer([long_text], max_length=1024, return_tensors="pt", truncation=True)

# Generate summary IDs directly from the model
summary_ids = model.generate(
    inputs["input_ids"],
    num_beams=4,
    max_length=120,
    min_length=50,
    early_stopping=True,
    no_repeat_ngram_size=3
)

# Decode the generated summary
summary_text = tokenizer.decode(summary_ids[0], skip_special_tokens=True)

print("Long Text Summary:\n")
print(summary_text)
```

```
Long Text Summary:

Artificial Intelligence has become a major area of research andinnovation. Organizations across ir
```

```
results = []

for t in texts:
```

```
    # Encode the input text
    inputs = tokenizer([t], max_length=1024, return_tensors="pt", truncation=True)

    # Generate summary IDs directly from the model
    summary_ids = model.generate(
        inputs["input_ids"],
        num_beams=4,
        max_length=70,
        min_length=25,
        early_stopping=True,
        no_repeat_ngram_size=3
    )

    # Decode the generated summary
    summary_text = tokenizer.decode(summary_ids[0], skip_special_tokens=True)
    results.append(summary_text)

print("\nAll Generated Summaries:")
for r in results:
    print("-", r)
```

```
All Generated Summaries:
- Companies use machine learning algorithms to analyze huge volumes of data and improve decision m
- Climate change is a major global challenge. Rising temperatures,melting glaciers, and increasing
- Software engineering involves designing, developing, testing, andmaintaining software applicatic
- A young girl discovered a hidden path that led to a mysterious ancient tree. The villagers belie
- Two colleagues were discussing their project during a meeting. One of them was worried about the
```